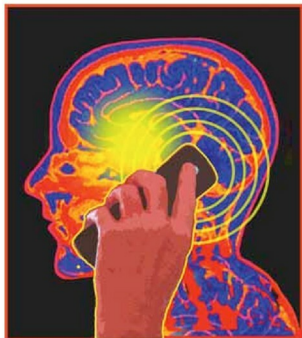


Cellphone Hazards And Recommendations

Mobile phones have been found to cause changes in the brain activity, reaction times and sleep patterns. The length of cellphone call affects the intensity of these symptoms. As it's not possible for us to stop using mobile phones altogether, precautionary measures are needed to avoid their ill effects



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Mobile phones use electromagnetic radiation in the microwave range, which may be harmful to human health. They communicate by transmitting radio waves through a network of fixed antennae called base stations.

Radio-frequency waves are electromagnetic fields, and unlike ionising radiation such as X-rays or gamma rays, cannot break chemical bonds nor cause ionisation in the human body. All radiations and fields of the electromagnetic spectrum that do not normally have sufficient energy to produce ionisation in matter are termed as

non-ionising radiation. These are characterised by energy per photon of less than about 12 electron volts (eV), wavelengths greater than 100 nanometres (nm) and frequencies lower than 3×10^{15} Hz.

In response to public and governmental concern, the World Health Organisation (WHO) started the International Electromagnetic Fields (EMF) Project in 1996 to assess the scientific evidence of possible adverse health effects of electromagnetic fields. Given the immense number of people who use mobile phones, even a small

increase in the incidence of adverse effects on health could have major public health implications. Because exposure to the RF fields emitted by mobile phones is generally more than a thousand times than from base stations, there is a greater likelihood of any adverse effect of handsets.

Health hazards

A large number of studies have been performed over the last two decades to assess whether mobile phones pose a potential health risk. An assessment was published in 2007 by the European Commission Scientific Committee on Emerging and Newly Identified Health Risks (SCENIHR). The research clearly

showed an increased risk of traffic accidents when mobile phones (either handheld or with a hands-free kit) are used while driving.

Electromagnetic interference is also a major concern. When mobile phones are used close to some medical devices (including pacemakers, implantable defibrillators and certain hearing aids), there is the possibility of interference with their operation.

Other health effects of using mobile phones include changes in the brain activity, reaction times and sleep patterns.

Radiation absorption. Part of the radio waves emitted by a cellphone is absorbed by the human head. The rate at which the radiation is absorbed by the human body is measured by the specific absorption rate (SAR), and its maximum levels for latest handsets have been set by governmental regulating agencies in many countries.

SAR is defined as the time derivative of the incremental energy absorbed by or dissipated in an incremental mass contained in a volume element of a given density. SAR values are heavily dependent on the size of the averaging volume.

In the USA, the Federal Communications Commission (FCC) has set an SAR limit of 1.6W/kg, averaged over a volume of one gram of tissue, for the head. In Europe, the limit is 2W/kg, averaged over a volume of 10 grams of tissue.

Measurement of SAR value is a complicated process and comprises a phantom human head, specially developed liquids, a robot fitted with a measurement probe and a mobile phone powered up to its maximum